

AMD Ryzen 9 9950X3D2 Dual Edition

Beyond gaming, into serious workloads

The AMD Ryzen 9 9950X3D2 is an unusual halo desktop CPU, not because it adds more cores, but because it doubles down on what has made AMD's X3D chips so interesting in the first place: stacked cache. Thanks to CoWoS! This is the first desktop processor to place second-generation 3D V-Cache on both Zen 5 CCDs, giving it 16 cores, 32 threads, and 208 MB of total cache on the AM5 platform. With the previous X3D SKUs, the story was all about gaming, but this time around AMD is talking about the processor more as a top-end chip for developers, creators, and other latency-sensitive workloads that can benefit from a very large on-chip memory pool. Is there a tradeoff? Well, of course there is. Compared with the standard Ryzen 9 9950X3D, the 9950X3D2 carries more cache and a higher 200W TDP, but a slightly lower maximum boost clock of 5.6 GHz.

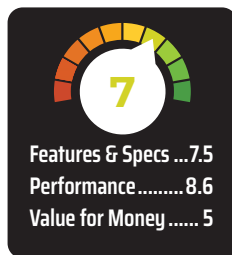
That's just 100 MHz lower. While AMD hasn't disclosed the price, we're seeing market prices of approximately Rs 98,499 which is quite ridiculous when you know that the AMD Ryzen 9 9950X3D goes for Rs 71,500. So, is it worth it? Does the extra cache make that much of a difference?

PERFORMANCE

Cinebench 2024

Cinebench 2024 is based on Maxon's Cinema 4D software, designed to evaluate a processor's performance in rendering complex 3D scenes. It tests single and multi-core capabilities, highlighting how efficiently a CPU handles multi-threaded tasks in rendering

₹1,25,000



workflows. The benchmark uses modern instruction sets and large datasets, providing an assessment of CPU performance in contemporary applications.

V-Ray

V-Ray is another benchmark that measures a processor's performance in rendering but it uses the V-Ray engine, widely used in visual effects, architecture, and design industries. It tests both CPU and GPU rendering capabilities, focusing on multi-threaded performance and handling complex calculations involved in ray tracing. CPUs with higher core counts and multi-threading technology tend to excel here as V-Ray efficiently utilises available threads.

AIDA64

AIDA64 Memory Benchmark measures the memory bandwidth and latency of a system, highlighting how quickly data transfers between the CPU and RAM.

It tests read, write, and copy speeds, providing insights into the efficiency of the memory subsystem, including RAM speed, timings, and memory controller performance. AIDA64 also has benchmarks to evaluate how well a processor can handle AES, ZLib and SHA3 encryption and decryption tasks.

Procyon AI Computer Vision

The UL Procyon AI Computer Vision benchmark measures the performance of AI inference engines to understand how processors handle machine-vision tasks using neural networks.

7 Game Average

While 3DMark is an excellent synthetic benchmark, it's not really indicative of real-world gaming performance and that's why we have to rely on popular AAA video games to assess how well the processor deals with contemporary video game engines. We revise the video games once every two years based on what's most popular as per Steam and we also maintain a mix between different game engines.

Core to Core Latency

Core-to-Core Latency measures the time it takes for data to transfer between different cores within a CPU. It highlights the efficiency of the processor's inter-core communication pathways and cache coherency mechanisms. An interesting insight is that lower core-to-core latency can improve performance in multi-threaded applications where threads need to synchronize or share data frequently. It can reveal architectural differences between CPUs, such as the impact of chiplet designs versus monolithic die layouts, affecting inter-core communication efficiency.

The AMD Ryzen 9 9950X3D2 has the same 16 Cores with two CCDs as the older 9950X3D and we can see that within the CCD, the latency for core-to-core communication is around 18-25 nanoseconds. And when switching to cores on the other CCD, the latency climbs into the range of 85-91 ns. This